Measuring the Diversity a Feed Actually Exposes

A standard, its attacks, and what it takes to verify it

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Abstract

A regulator seeking to constrain a recommender algorithm needs a quantity it can observe without access to the platform's code. This note proposes one — the **Exposed Diversity Index** (EDI), the entropy of the items served, projected onto a declared catalogue of viewpoints and weighted by the attention each rank commands — and then subjects it to the attacks it must withstand to serve as a standard.

Three of those attacks succeed against the naive formulations, and each forces a correction. A floor bearing on a feed's *labels* is saturated at zero cost: a platform able to decouple label from content obtains full marks for strictly zero content diversity. Rao's quadratic entropy, the first replacement considered, has a *bimodal* constrained optimum: it would prescribe the very polarisation it claims to measure. And any rank-blind measure is satisfied by *burying* the divergent items — a platform certified at 0.70 exposes only 0.36 of real diversity, and closing the loophole doubles the engagement cost.

Verifying such a standard on real data requires knowing *exposure*. We give an exact within-feed test that decides whether a recommender log permits any exposure correction at all, and an estimator of position-bias severity. Applied to three public logs, they establish that MIND does not preserve the served order — the estimator returns five incompatible severities there — while Baidu-ULTR does ($\hat{\eta} = 1.10 \pm 0.09$), and the Open Bandit Dataset permits this work's only confrontation with a ground truth: the value of a never-deployed policy is recovered to within 2.5%, against 31.6% error for the naive estimate.

No public dataset carries both the served rank and an interpretable viewpoint label. We therefore specify, under Article 40 of the *Digital Services Act*, four aggregate tables containing no personal data, verified to recompute the measurements identically for a hundred times fewer rows than a raw log.

Part II presents the thermodynamic model the index came from, and §15 reports what remains of it: the quantum decoherence analogy that started this work survives none of its transfers. All code, tests and figures are reproducible in containers.

Part I

The measurement

1 The metrological problem

Regulating a recommender algorithm presupposes an observable quantity. Counting problematic items will not do: the indicator lags, it is gameable, and it requires ruling on the substance of every item. The quantity adopted here is of a different nature — it bears on the *shape* of exposure, not on the content of messages:

> What share of the attention served to a reader goes to each of the viewpoints the platform could have served?

The question has three virtues: it can be measured from outside, it requires no judgement on the truth of any item, and it admits a threshold expressed as a percentage, hence transferable across platforms.

It also has three traps, and this note is organised around them.

1. **A label is not an item.** Measuring the diversity of *announced categories* leaves the platform to choose what is announced.
2. **A composition is not an exposure.** A feed may contain the required diversity and place it where nobody looks.
3. **A logged click is not a preference.** Observed clicks were produced by the platform's ranking, not by the one under evaluation.

Each trap is measurable, and each was measured. What survives all three is the retained definition.

2 The exposed diversity index

Given a feed of n positions, a reference catalogue of k viewpoints declared by the regulator, and an attention discount w_R attached to rank R, the *exposed* distribution is

$$q_i = \frac{\sum_{R=1}^{n} w_R \mathbb{1}[\text{the item served at rank } R \text{ falls in bin } i]}{\sum_{R=1}^{n} w_R}, \tag{1}$$

and the index is its normalised Shannon entropy:

$$\mathrm{EDI} = \frac{H(q)}{\log_2 k}, \qquad H(q) = -\sum_{i=1}^{k} q_i \log_2 q_i. \tag{2}$$

Three choices distinguish (1) from the entropy one would write spontaneously, and each follows from an attack that succeeded (§3).

Choice	**Reason**	**Without it**
q bears on the items served, projected onto the catalogue's bins	a label is chosen, an item is observed	the index reaches 1.000 for zero content diversity, at zero cost
each rank is weighted by w_R	a reader consults the first item far more often than the last	the standard is satisfied by burial: 0.70 certified, 0.36 exposed
the denominator $\log_2 k$ comes from the declared catalogue	comparing platforms requires the same unit	a closed feed shows one modality, the denominator degenerates and the index flatters

Attention discount. The results below use $w_R = 1/R$, the reciprocal-rank discount standard in ranking evaluation. The choice of w belongs to the regulator, as does the catalogue; what matters here is that it be *declared*, since it is what makes the standard non-circumventable by order.

Associated control quantity. The gap between (2) computed on the feed's composition (w_R constant) and on its exposed distribution ($w_R = 1/R$) is zero for a platform that does not relegate, and grows with burial. It is the only quantity in this work that compares a measure *with itself*, and hence the only one that thresholds without an external reference.

How to publish the figure. A normalised entropy is not a diversity: it is not linear in what we mean by "twice as diverse". Its conversion into an **effective number of viewpoints**, ${}^1D = k^{\mathrm{EDI}}$, is [Jost, 2006]: a floor of 0.70 on a catalogue of four is 2.6 effective viewpoints, and the 0.44 actually exposed by the burying feed is 1.9.

Protocol. The unit is the item *served*, not the item consulted: the platform is audited, not the reader. The index is computed per reader over a sliding window and then aggregated; the regulatory quantity is the *share of the population below the threshold*, not the mean, since a satisfactory mean can hide an entirely enclosed minority.

3 The adversarial test

A standard is judged by what it forces a rational adversary to do. We therefore place the platform in the role it will occupy: maximising engagement *subject to* the constraint, rather than complying in good faith.

Method. A feed has n positions to fill from a catalogue of k viewpoints, hence k^n possible feeds. For $n = 8$ and $k = 4$, all 65,536 feeds are *enumerated*: the optimum obtained is exact, not the output of a heuristic. This is not a luxury. These are negative results — standards that fail — and an optimum missed by a solver would look exactly like a standard that holds.

3.1 A label-based standard saturates at zero cost

If the platform controls the label independently of the item, the entropy of labels stops constraining the entropy of items. Optimisation confirms it without nuance: under full decoupling the platform obtains an index of 1.000 — full marks — for strictly zero content diversity, without giving up a point of engagement. One need not go that far: at half decoupling, the constraint retains only 36% of its force.

The consequence is direct: **the measure must bear on the items served**. It calls for a complementary diagnostic, the *signature excess* — the gap between two indices computed on the same feed, relative to what an honest catalogue would show at the same index — which is zero for an honest platform and grows with decoupling.

3.2 The first replacement prescribed polarisation

Rao's quadratic entropy [Rao, 1982] — the mean distance between two items drawn from the feed, known in recommendation as *intra-list distance* — appeared to answer the objection: it sees items, not labels. Yet its engagement-constrained optimum is **bimodal**: at a given required diversity, the cheapest way to supply it is to serve two extreme blocks and empty the centre. A standard built on it would prescribe the polarisation it claims to measure. The defect is known in the diversification literature [Ohsaka and Togashi, 2023]; it is recovered here as a direct consequence of constrained optimisation.

The retained floor therefore bears on an *entropy*, computed on the items served projected onto the catalogue's bins — a measure whose constrained optimum is spread rather than bimodal.

3.3 A rank-blind standard is circumvented by burial

The measures above bear on the feed's *composition.* Retested on *ordered* feeds, all four candidate diversity measures are circumvented the same way: serve the divergent items at the last ranks, where attention no longer goes.

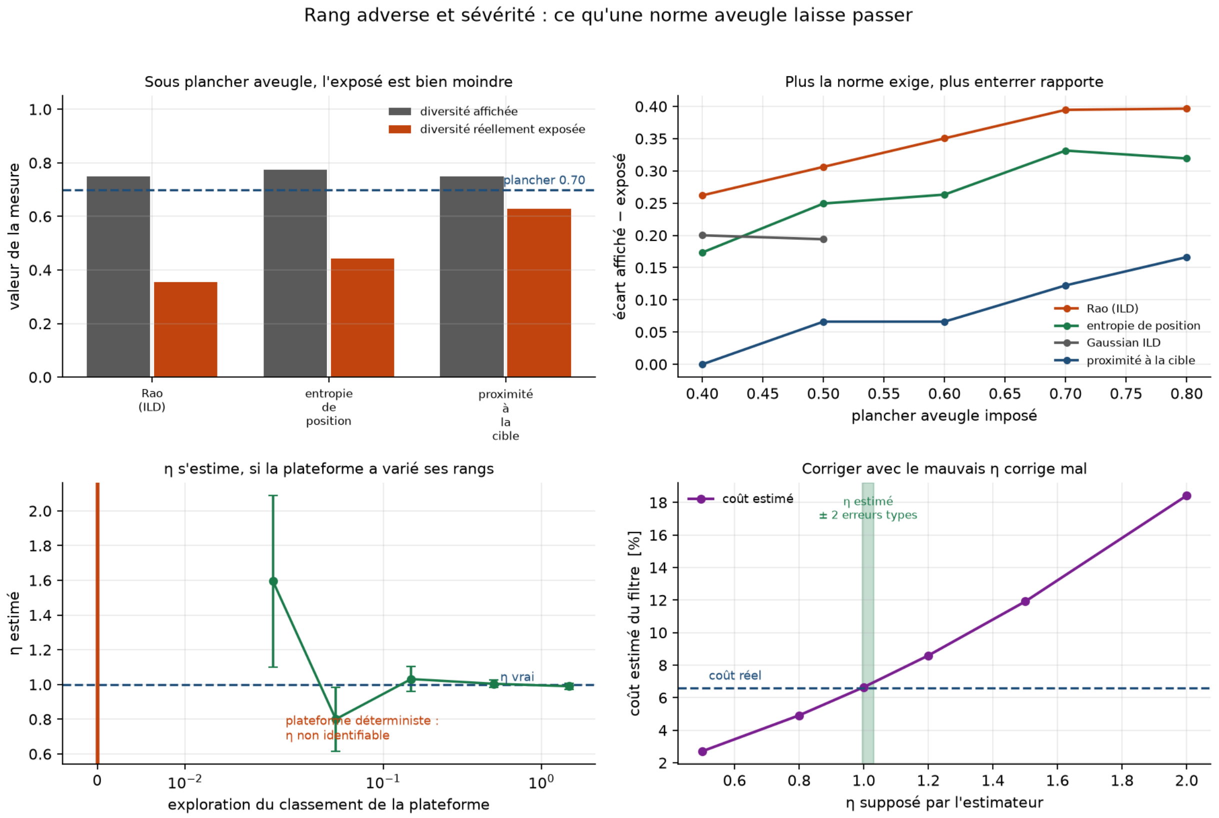

Figure 1: Displayed versus exposed diversity under a rank-blind floor; the gap by floor level; position-bias severity recovered by fixed effects, and the exploration threshold below which it ceases to be identifiable; the cost of a posited η.

Measure	**Cost**	**Displayed**	**Exposed**	**Cost if rank-aware**
Rao's entropy	8.2%	0.750	**0.355**	18.9%
Position entropy	10.7%	0.774	**0.443**	20.9%
Gaussian ILD	floor unattainable			
Target proximity	5.9%	0.750	**0.628**	10.6%

The optimal feed always has the same shape: items of the preferred viewpoint on top, divergent ones relegated. **A platform certified at 0.70 exposes only 0.36.** A rank-aware floor closes the loophole and *doubles* the engagement cost: a standard that cost no more would close nothing.

The gap grows with the requirement — 0.262 at a floor of 0.40, 0.395 at 0.70 for Rao's entropy: the more diversity a rank-blind standard demands, the more profitable burial becomes.

4 Measuring exposure

Everything above presupposes knowing what attention goes to which rank. The standard model posits an examination probability decreasing with rank [Joachims et al., 2017],

$$e(R) = R^{-\eta}, \tag{3}$$

whose severity η was, in the first version of this work, *posited*. It can be estimated.

4.1 Estimation by item fixed effects

Under (3), click probability factorises as $P(\text{click} \mid i, R) = g(i)\, R^{-\eta}$, hence

$$\log \mathrm{CTR}(i, R) = \log g(i) - \eta \log R. \tag{4}$$

Relevance $g(i)$ is an item fixed effect: it is not estimated but eliminated by within-item centring. What remains is the only variation that identifies η — that of the *same item seen at different ranks*. This is the simplest form of intervention harvesting [Agarwal et al., 2019]: it requires no experiment, only that the platform did not always rank the same items in the same places.

On simulated data the estimator recovers η from 0.4 to 1.6. Under a deterministic policy it *refuses* to answer rather than invent a figure: no item changes rank, and the severity is not in the data.

Positing η instead of estimating it is not innocuous: where the true cost of a diversity filter is 6.6%, a posited η of 0.5 understates it by 59% and a posited η of 2.0 overstates it by 179% — the very order of magnitude of the bias one claimed to correct.

4.2 The check that must come first: exchangeability

The refusal described above catches the visible case — no rank variation — and misses the invisible one: *abundant and artificial* variation. What is needed is a check that counts not the variation but the *informativeness* of the order.

For a feed of length L carrying k clicks, let $u_R = (R - 1/2)/L$ be the normalised rank. Under the hypothesis that clicks are indifferent to position *given the feed*, the clicked positions are a draw without replacement among the L positions, with exactly known expectation and variance:

$$\mathbb{E}\Big[\sum_{\text{clicks}} u_R\Big] = k\,\bar{u}, \qquad \mathbb{V}\Big[\sum_{\text{clicks}} u_R\Big] = \frac{k(L-k)}{L-1}\,\sigma_u^2. \tag{5}$$

Conditioning on the feed immunises the test against the mixture of feed lengths, the mean quality of a feed's items, and its reader's appetite for clicking. Position bias concentrates clicks at the top: it makes the standardised deviation *negative*, so the sign of the rejection is itself a verification.

A test that rejects nothing says nothing until one knows what it would reject. Calibration is done by simulation at identical feed structure: on MIND's, the test detects $\eta = 0.02$ at twelve standard deviations, and the minimum detectable severity is $\eta \approx 0.004$.

5 Evaluating a re-ranking

Measuring a feed is one thing; estimating what a different ranking would cost is another. The available clicks were produced by the platform's policy, not by the one under evaluation.

Replay is wrong, by a considerable margin. Replaying logged clicks on a re-ranked feed is off by 201% at the median, up to 851%, and the sign of the error is not guaranteed: across sixty draws at identical configuration it overstates in fifty-six cases and understates in four. A naive figure is therefore not even an upper bound.

The correction and its price. Inverse propensity weighting is unbiased provided the logging policy is known and gives every action the target policy might choose a chance. Its price is variance, hence the usual variants — self-normalisation [Swaminathan and Joachims, 2015], weight clipping — and the diagnostic that must accompany any figure: the **effective sample size**, which states how many observations actually carry the estimate.

A confrontation with ground truth. The Open Bandit Dataset [Saito et al., 2020] contains two buckets served in parallel by two different policies, and publishes the true propensities. One can therefore estimate the value of the uniform policy *from another policy's data alone*, and compare it with its value measured where it actually served.

Estimator	**Value**	**Error**
Ground truth (measured, 452,949 impressions)	0.005124	—
Naive (observed click rate)	0.006743	+31.6%
IPS	0.005253	**+2.5**%
Self-normalised	0.004768	−7.0%
IPS clipped at 10	0.004473	−12.7%

The correction works. The diagnostic forbids celebrating it: the effective sample size is 1,513 for 4,077,727 impressions, i.e. 0.04%. The estimate is unbiased and rests on the equivalent of fifteen hundred observations; that the error lands at 2.5% owes as much to luck as to method. This is the real case that justifies the rule: publish the effective size next to the figure.

6 Three public logs

The instruments of §4 and §5 decide whether a dataset permits the measurement. Applied to three public logs, they give three different answers.

6.1 MIND: an order that is not one

The reference dataset for news recommendation [Wu et al., 2020] shows ideal rank coverage: a median item appears at sixteen distinct positions. Yet test (5) detects nothing — $z = +0.12$ ($p = 0.91$) across 156,965 feeds, replicated at $z = +0.28$ on the second split — in a dataset where it would detect $\eta = 0.02$ at twelve standard deviations. The recorded order is indistinguishable from a shuffle.

The click-rate curve by position nonetheless falls from 0.108 to 0.038 and fits $\hat{\eta} = 0.39$: a *composition artefact*, deep positions existing only in long feeds, whose click rate per item is mechanically lower. At fixed feed length the slope is zero and changes sign from one length to the next.

Finally, estimator (4) accepts the dataset, declares itself identifiable, and returns a severity that is an increasing function of a mere nuisance threshold: −0.131 at five impressions, +0.053 at twenty, +0.255 at a hundred, each with a standard error below 0.007. **Five confident and incompatible figures from one dataset.**

Shuffling does not debias the clicks: they were produced under the real order and carry its bias in full. What the shuffle removes is the *rank*, the only regressor that would allow correction. On a simulated log of true severity 1.00, the estimate moves from 1.003 to −0.003 after shuffling, at identical clicks; evaluating a re-ranking becomes vacuous by construction.

6.2 Baidu-ULTR: the positive control

On a slice of 524,164 documents served across 64,200 search sessions [Zou et al., 2022], the same test rejects at $z = -205.7$ ($p < 10^{-12}$) and on the side theory prescribes. Severity is $\hat{\eta} = 1.10 \pm 0.09$ by document fixed effects, against 1.49 by aggregate fit — the gap being the quality confounder, the platform placing the best documents on top. Coverage remains thin: 55 documents carry the estimate, out of 444,709, because the same URL rarely reappears.

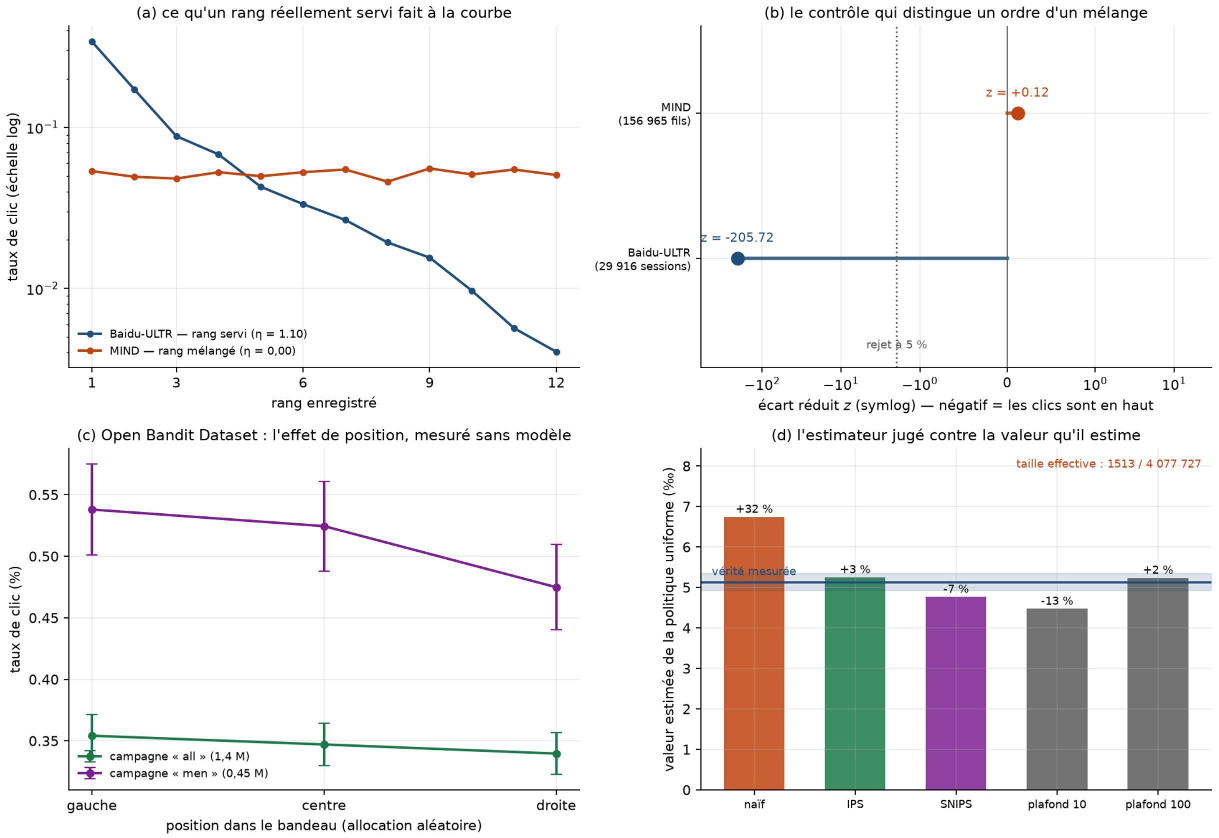

Figure 2: What a genuinely served rank does to the click-rate curve; the exchangeability test on MIND and on Baidu-ULTR; the position effect measured without a model on the Open Bandit Dataset's random bucket; the counterfactual estimator judged against the value it estimates.

6.3 Open Bandit Dataset: position measured without a model

The random bucket allocates items to positions by chance: the click-rate difference between positions is a *causal* effect, with no examination model to posit. On a three-thumbnail horizontal banner it is $\hat{\eta} = 0.037$ ("all" campaign, 1.37 million impressions) to $\hat{\eta} = 0.105$ ("men" campaign).

η **is therefore a property of the surface, not a constant**: an order of magnitude separates a vertical results page from a three-thumbnail banner. Transporting one to the other is exactly the error whose cost §4 quantifies.

6.4 What these three logs do not permit

None carries both the *served rank* and an *interpretable viewpoint label*. MIND has editorial categories without the rank; Baidu-ULTR has the rank without labels; the Open Bandit Dataset has rank, propensity and three categorical attributes, but anonymised — and a diversity computed over hashes says nothing, the reference catalogue being a political declaration.

The index of §2 has therefore never been measured on a real feed. This is not a gap in method: it is a gap in data.

7 What to ask for

Article 40 of the *Digital Services Act* [European Union, 2022], made operational by Delegated Regulation (EU) 2025/2050 [Commission européenne, 2025], opens platform data to vetted researchers. A request is admissible only if it is *necessary and proportionate*. "Give us your logs" is neither: it demands personal data the analysis has no need of, and it hands over the easiest ground for refusal — service security and trade secrets.

We therefore specify the request as **four aggregate tables**:

1. **feed profiles** — for each served feed shape (the list of ranks it occupies) and each click count, the number of feeds;
2. **clicks by rank** — for each profile, each feed click count and each rank, the number of clicks;
3. **(item, rank) cells** — impressions and clicks, and the service propensity if the platform knows it;
4. **exposure by (rank, viewpoint)** — impressions and clicks, over the catalogue declared by the regulator.

No row designates a reader: a feed appears only by its shape and by the number of clicks it received.

Sufficiency verified, not asserted. Test (5) is recomputed from tables 1 and 2 alone to within 3×10^{-12} on MIND as on Baidu-ULTR; severity (4) is recomputed *exactly* from table 3; both diversity measures from table 4, which therefore suffices to observe burial — at identical composition, a feed compliant with the rank-blind floor exposes only 0.160 of attention to minority viewpoints, against 0.375 in composition.

Proportionality quantified. What is asked weighs 5,543 rows for the 524,164 documents served in the Baidu-ULTR slice (ratio 95) and 57,906 rows for MIND's 5,843,444 served items (ratio 101).

The confidentiality threshold is not neutral. Suppressing low-count cells protects readers and moves the estimates: going from 5 to 20 impressions per cell moves the estimated severity from 1.10 to 1.40, i.e. +27%, because rare cells are those of deep ranks. The threshold must be fixed in the request and published with the result.

Two simpler aggregation keys were discarded because they were wrong without being visible: indexing feeds by length rather than by rank profile — wrong as soon as a surface skips ranks — and omitting the feed click count, which prevents excluding fully clicked feeds and shifts the standardised deviation from −205.7 to −203.9. Both produced figures of the right order of magnitude.

8 A filter that optimises the index

A classical recommender orders items by estimated relevance. The filter proposed here adds a term:

$$S(i,c) = \mathrm{Relevance}(i,c) + \mu\,\Delta H(i,c), \qquad \mu \geq 0, \tag{6}$$

where ΔH is the change in index item i would bring to feed c. At $\mu = 0$ the ranking is *identical* to an engagement filter's: the transformation is a parameterised addition, not a redesign. The parameter μ follows an annealing schedule — rising when the index falls below the threshold, falling once it is restored — so that intervention is intermittent.

Status. This filter is validated against its own model, with synthetic relevance and four viewpoints. Its real cost in perceived relevance is not measured, and cannot be on public data as things stand (§6). What bounds the discussion is the figure of §3: a rank-aware floor costs 10.6 to 20.9% of engagement depending on the measure, on eight-slot feeds enumerated exhaustively.

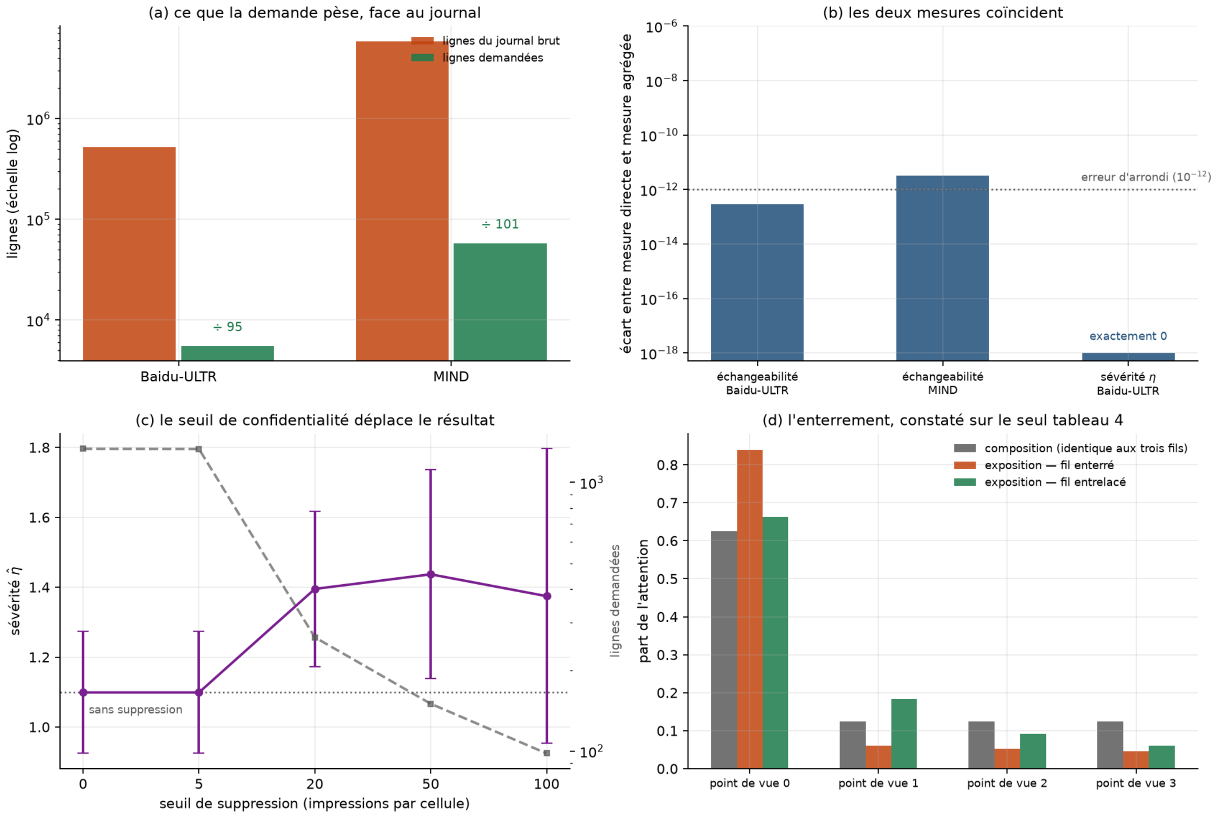

Figure 3: What the request weighs against the log; the gap between direct and aggregate measurement; the effect of the confidentiality threshold on estimated severity; burial observed from the exposure table alone.

8.1 What it is worth, compared with baselines

A filter is not judged against relevance ranking, from which it differs by construction, but against what a practitioner would do without theory. We therefore compare it with four baselines — serving viewpoints in turn, *maximal marginal relevance* [Carbonell and Goldstein, 1998], a Boltzmann draw on relevance, and random selection — and above all with the **exact frontier** of the (exposed diversity, engagement) plane, obtained by enumerating the 151,200 ordered arrangements of six items drawn from a pool of ten.

Across 150 pools, at a fixed floor, the median shortfall is as follows.

Method	0.50	0.60	0.70	0.80	0.90
Round robin	0.0%	0.0%	0.0%	3.8%	4.8%
MMR	0.0%	0.0%	0.0%	**0.0%**	**0.2%**
Boltzmann	0.0%	0.0%	0.1%	1.2%	3.7%
Filter (6)	0.0%	0.0%	0.0%	**0.0%**	**1.0%**
Random	9.3%	9.8%	11.0%	12.9%	19.1%

The filter therefore holds the frontier — which was not a given, random selection reaching the same diversities while leaving a fifth of the engagement on the table — but **it is not alone there**. A heuristic published in 1998 holds it too, and beats the filter head to head at floor 0.80. The filter only stands out at the high requirement: at floor 0.90, MMR finds no compliant setting in 47% of pools, against 12% for the filter, because the latter optimises the constrained quantity where MMR optimises a proxy that saturates.

The price of the standard depends on the reader. The comparison above draws relevance independently of viewpoint. Varying that *alignment*, the cost of the 0.90 floor moves from 3.8% of engagement — a reader whose interests cut across viewpoints — to 17.1% — a reader whose preference *is* a viewpoint. The standard therefore costs most exactly where it serves most, which predicts where the objection will fall. This dependence reconciles the figures of §3, where relevance was attached to the viewpoint, with those of this paragraph.

Part II
The background model

What precedes stands without what follows. The index, its attacks, the exchangeability test and the counterfactual estimators depend on no hypothesis about opinion dynamics: they are measurements on served feeds and observed clicks.

This part presents the model the index *came from*. It has two functions, and only one is scientific. The first is to account for provenance: the quantity adopted in §2 was not picked at random among possible diversity measures, it comes from a formalism that describes opinion as a system with a temperature. The second is to state what that formalism is worth once measured — and §15 concludes that the analogy which suggested it survives none of its transfers.

No result in Part I rests on this part.

9 Entropic formalism

9.1 Two entropies, one quantity

The von Neumann entropy [von Neumann, 1932]

$$S(\rho) = -\operatorname{Tr}(\rho \ln \rho) \tag{7}$$

vanishes for a pure state. It is computed by diagonalisation: the eigenvalues of ρ form a classical distribution to which the Shannon entropy [Shannon, 1948] applies,

$$H(X) = -\sum_i p_i \log_2 p_i. \tag{8}$$

This is precisely what licenses the bridge between the two disciplines: von Neumann entropy *is* the Shannon entropy of the mixture revealed in the eigenbasis.

9.2 A necessary clarification

One common and incorrect formulation must be set aside at once. The evolution of a *closed* quantum system is unitary, so its von Neumann entropy is constant. What grows under decoherence is the entropy of the *reduced subsystem*, obtained by tracing over environmental degrees of freedom.

A clarification follows that strengthens the analogy rather than weakening it: a coherent superposition is a *pure* state of zero entropy. It is therefore not the multiplicity of possibilities that produces disorder, but the loss of coherence between them. The social counterpart is more accurate this way: a population where everyone keeps several opinions open is not disordered — disorder arises from contact with an environment that fixes positions.

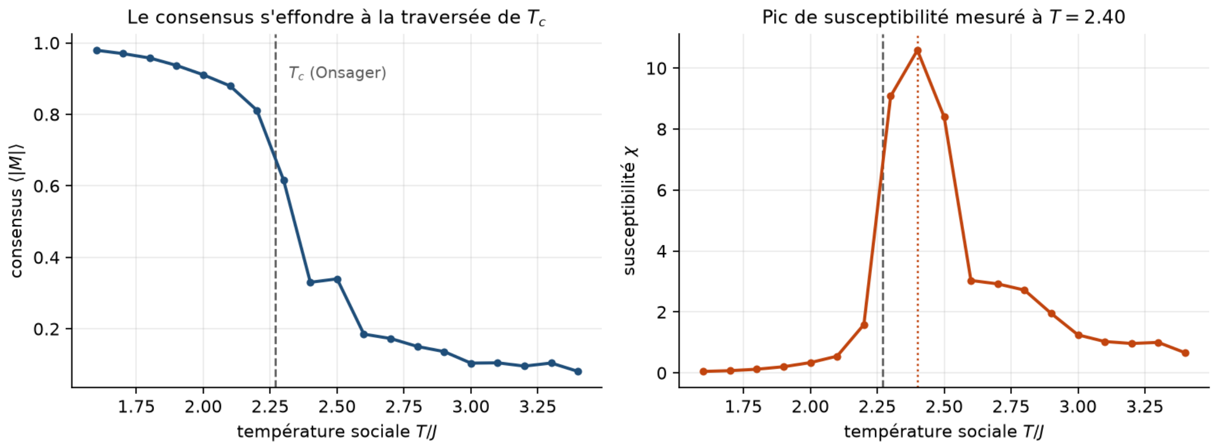

Figure 4: Consensus and susceptibility as functions of social temperature (Monte Carlo, 24×24 lattice). The susceptibility peak locates the transition. Its upward shift relative to Onsager's value (10) is a finite-size effect, not an implementation error. Figure captions are in English; notebook prose is in French.

10 Thermodynamic dynamics

10.1 The Ising model and social temperature

Each individual carries a binary opinion $s_i \in \{-1, +1\}$ and the population follows the Gibbs-Boltzmann distribution [Ising, 1925]:

$$P(\sigma) = \frac{1}{Z} \exp\left(\frac{J \sum_{\langle i,j \rangle} s_i s_j + H \sum_i s_i}{k_B T} \right), \tag{9}$$

where J measures local conformity, T is the *social temperature* (agitation, irrationality, openness to fluctuation) and H the external media field.

The model's value lies not in the analogy but in a verifiable property: the existence of a critical temperature. In two dimensions its exact value is known [Onsager, 1944]:

$$\frac{T_c}{J} = \frac{2}{\ln(1 + \sqrt{2})} \approx 2.269. \tag{10}$$

This is the only point in this work where an exact theoretical prediction, independent of our sociological assumptions, can validate the implementation. Figure 4 recovers it numerically; the residual gap is a finite-size effect on a 24×24 lattice.

10.2 Free energy

The system minimises its Helmholtz free energy $F = E - TS$, where E represents the tension of disagreement and TS the dispersive force. As N grows, configurational entropy grows with it and the TS term eventually dominates E.

10.3 Social hysteresis

This is the most directly actionable result. Below T_c, switching off the media field ($H \to 0$) does not return magnetisation to zero: the interaction term $-J \sum s_i s_j$ takes over and the group sustains the belief through internal cohesion.

The cycle reads in three stages, which are those of a media crisis: injection of a massive field, official retraction, then the need for a counter-field $-H$ exceeding the group's coercive field. Figure 5 shows the cycle area decreasing with social temperature and vanishing above T_c.

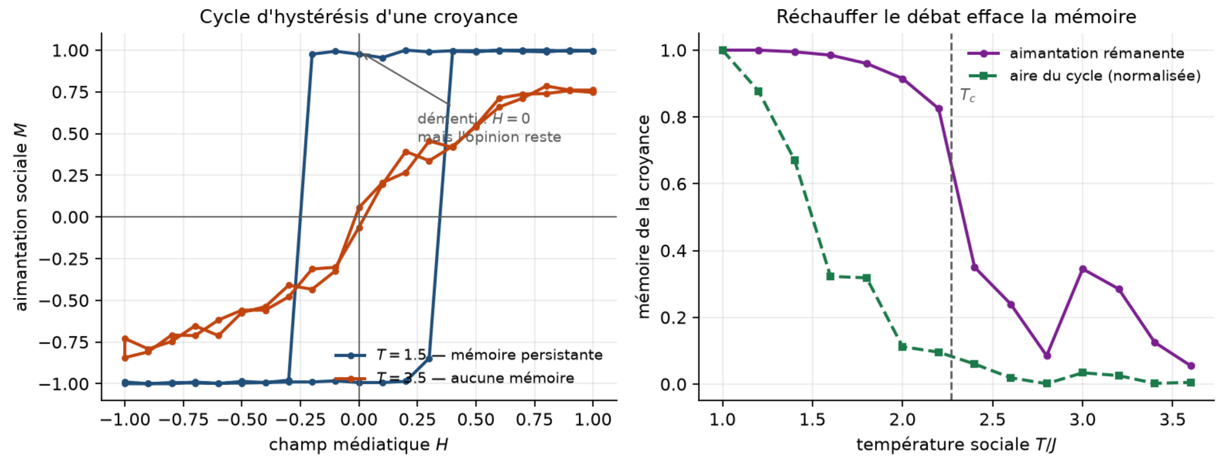

Figure 5: Left: hysteresis cycles below and above T_c; in the former, opinion remains magnetised while the field is zero. Right: decay of belief memory with social temperature — the basis for the thermal-noise recommendation.

Practical consequence: passive debunking does not work. Restoring informational neutrality leaves the belief in place; only active counter-information, or a rise in social temperature, dissipates it.

11 Size effects: consensus time

The voter model [Clifford and Sudbury, 1973, Holley and Liggett, 1975] isolates pure social contagion, without temperature: at each interaction an individual adopts a randomly chosen neighbour's opinion. It yields exact scaling laws for consensus time, measured in sweeps (N elementary interactions).

Measurements (figure 6) give an exponent ≈ 1 in mean field and ≈ 2 on a ring. **A densely connected network therefore converges faster than a geographic neighbourhood.**

This invalidates a frequent argument, that the small-world structure of social networks [Watts and Strogatz, 1998] prevents consensus. Connectivity does not fragment: it accelerates and amplifies, in whatever direction the field gives it. The causes of fragmentation are the *directional bias* of algorithmic micro-fields and the *homophily* that compartments the graph [Pariser, 2011]. The regulatory consequence is clear: the useful lever acts on the field, not on the number of links.

12 The landscape of public opinion

12.1 The equation and its drift

At macroscopic scale, collective opinion $x \in [-1, 1]$ obeys a Fokker-Planck equation [Risken, 1989]:

$$\frac{\partial P(x,t)}{\partial t} = -\frac{\partial}{\partial x}[A(x)P(x,t)] + \frac{\partial^2}{\partial x^2}[B(x)P(x,t)]. \tag{11}$$

The quadratic potential $V(x) = -\frac{J}{2}x^2 - Hx$, which comes naturally to mind, can produce *no* phase transition: the corresponding exponential is convex, hence maximal at the extremes at any temperature. The missing term is the entropy of mixing — the number of individual configurations compatible with a mean opinion x. Adding it recovers the Helmholtz free energy:

$$f(x) = \underbrace{-\frac{J}{2}x^2 - Hx}_{E} + T\underbrace{\left[\frac{1+x}{2}\ln\frac{1+x}{2} + \frac{1-x}{2}\ln\frac{1-x}{2}\right]}_{-S}, \tag{12}$$

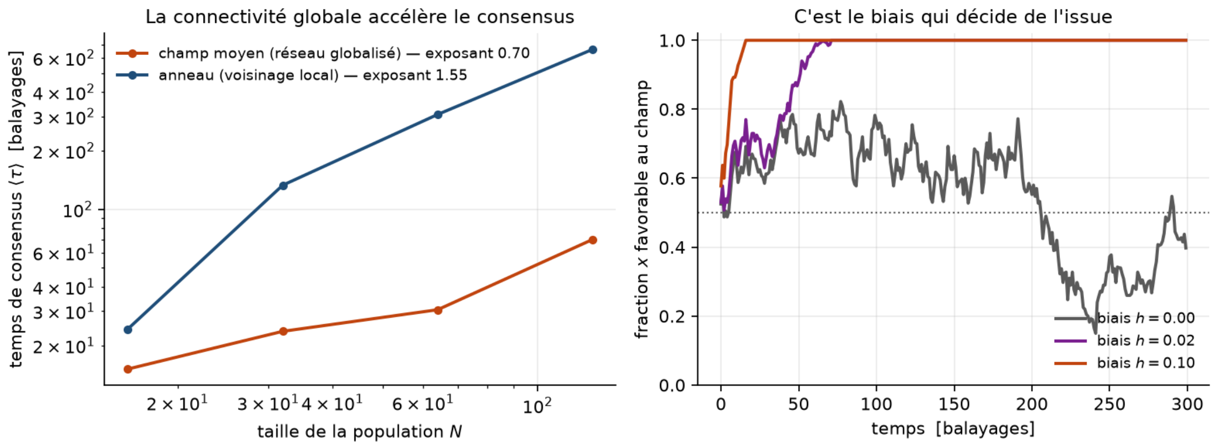

Figure 6: Left: consensus time versus population size, log-log, for two topologies. Right: trajectories of the field-favouring fraction for three bias intensities — it is bias, not connectivity, that decides the outcome.

giving the drift

$$A(x) = -f'(x) = Jx + H - T\,\mathrm{artanh}(x). \tag{13}$$

The entropic restoring term bounds the dynamics within $(-1, 1)$ and yields a mean-field critical temperature $T_c = J$. The form $A(x) = Jx + H$ is its linearisation near $x = 0$.

12.2 Diffusion and the effect of N

The diffusion term reads

$$B(x) = \frac{k_B T(1 - x^2)}{N}, \tag{14}$$

the factor $(1 - x^2)$ switching off noise at unanimity.

The $1/N$ factor requires careful reading, since it appears to contradict the initial hypothesis: the larger the population, the *less* noisy its macroscopic variable. There is no contradiction, but two distinct scales. Total configurational entropy is extensive and grows as N; fluctuations of the mean x decay as $1/N$. The defensible thesis is therefore:

> A large population does not become noisy, it becomes *rigid*. Its size does not agitate it; it deprives it of stochastic plasticity — and it is that rigidity which makes polarisation irreversible, since a system without noise can no longer leave the potential well it has fallen into.

This reformulation is stronger than the initial hypothesis: it explains irreversibility, which the "entropy pump" metaphor did not.

12.3 Three regimes

The equilibrium distribution takes the large-deviation form $P_{\mathrm{stat}}(x) \propto \exp(-Nf(x)/T)$, where the factor N makes the penalty more severe the larger the population. Three regimes follow, obtained by varying two parameters (figure 7): a fluid society ($T > T_c$), a polarised one ($T < T_c$, $H = 0$) and a manipulated one ($T < T_c$, $H > 0$). In the last case the peak on the field's side overwhelms the other: the "false consensus" is not an added assumption, it is the tilted well.

The density of moderates decays exponentially with N. In a large population below T_c, moderation is not a minority position: it is statistically inaccessible.

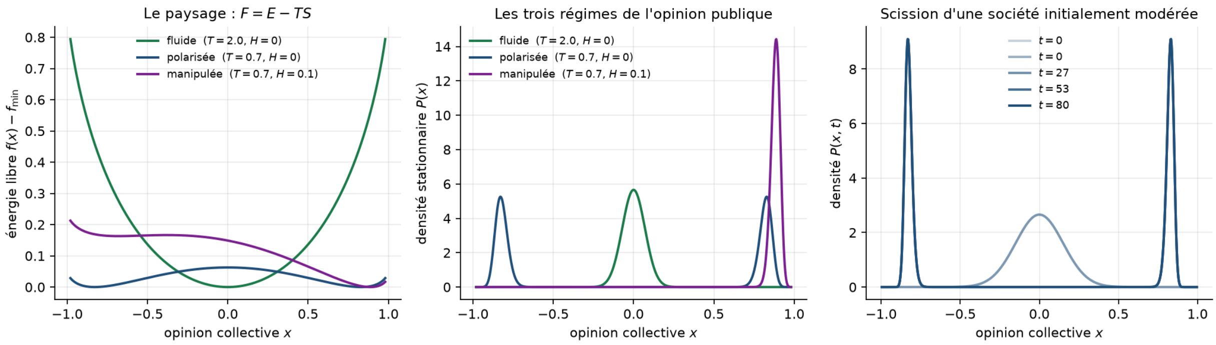

Figure 7: Free-energy landscape (12), the associated stationary distributions, and the temporal splitting of an initially moderate society. The solver uses finite volumes with zero flux at the boundaries: probability mass is conserved to 10^{-15}, which distinguishes a genuine collapse of moderation from numerical leakage.

12.4 Barrier crossing

Moving from one bloc to the other without passing through moderation is thermally activated barrier crossing, with rate $\propto e^{-\Delta V/k_BT}$ [Kramers, 1940], and not tunnelling — which has no classical counterpart. The distinction is not cosmetic: Kramers' law supplies a temperature dependence, hence a testable prediction.

13 Algorithmic resonance kinetics

A piece of false information and a recommender algorithm form a positive feedback loop analogous to the Larsen effect. Visibility $V(t)$ obeys

$$\ddot{V} + (\lambda - \gamma\alpha\,\sigma(V))\dot{V} + \omega_0^2 V = \xi(t), \tag{15}$$

where λ is natural damping (forgetting), γ the algorithmic gain, α the content's emotional charge, and $\sigma(V) = 1/(1 + (V/V_{\text{sat}})^2)$ a saturation factor expressing the finiteness of available attention.

The instability criterion is

$$\boxed{\gamma\alpha > \lambda} \tag{16}$$

Beyond it, effective damping becomes negative: the system accumulates energy instead of dissipating it. Figure 8 confirms that the measured threshold coincides with $\gamma^\star = \lambda/\alpha$.

Without saturation the unstable solution diverges as $e^{(\gamma\alpha-\lambda)t}$, which is exact but unusable: infinite visibility does not exist and no threshold is calibratable on a divergent model. With saturation the system settles into a limit cycle — a self-sustaining media oscillation of finite amplitude, matching the observed behaviour of established disinformation.

One point deserves emphasis for regulation. At *uniform* gain, a more emotional item crosses the threshold where a factual one does not. A platform therefore need not favour disinformation in order to amplify it selectively: the bias is mechanical. This is what makes measuring γ more relevant, and more enforceable, than auditing editorial intent.

14 Empirical calibration of the amplification ratio

Criterion (16) long remained a formal inequality: nothing indicated where real systems sit. We estimate it here on public attention series.

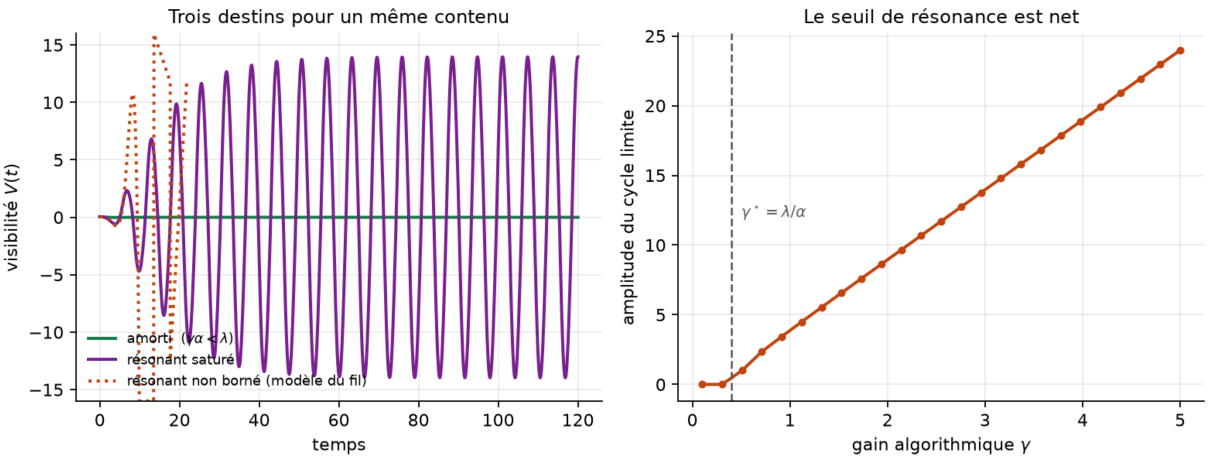

Figure 8: Left: three fates for the same content — damped, bounded resonant, unbounded resonant. Right: limit-cycle amplitude versus algorithmic gain; the measured threshold coincides with $\gamma^\star = \lambda/\alpha$.

14.1 Identification

For an isolated, non-oscillatory attention spike, (15) reduces to its first-order form $\dot{V} = \gamma\alpha\,\sigma(V)V - \lambda V$, which separates into two measurable regimes: during the *rise*, saturation is inactive and $V \propto e^{(\gamma\alpha-\lambda)t}$; during the *decay*, amplification is switched off and $V \propto e^{-\lambda t}$. Two log-linear regressions therefore give

$$\lambda = r_{\text{down}}, \qquad \gamma\alpha = r_{\text{up}} + r_{\text{down}}, \qquad \frac{\gamma\alpha}{\lambda} = 1 + \frac{r_{\text{up}}}{r_{\text{down}}}. \tag{17}$$

14.2 Data

The series come from the Wikimedia pageviews API, filtered to the `user` agent to exclude robots, over a *pre-registered* corpus of 24 subjects split into two emotional registers — accusation or scandal on one side, unscheduled discovery on the other. Since Wikipedia has no recommender algorithm, the estimated γ is an *ecosystem* gain rather than a platform's ranking function.

14.3 Results

Nineteen episodes are exploitable. The ratio lies between 1.5 and 12, with a median of 2.5 to 4.2 depending on the estimator (figure 9). Three findings, two of them negative.

The sign criterion is empty. The ratio exceeds 1 in every episode and under every estimator. This is not a property of the ecosystem but a *tautology of the procedure*: an observable episode necessarily went through a growth phase, so $r_{\text{up}} > 0$. The recommendation to "prohibit configurations where $\gamma\alpha > \lambda$" is consequently inapplicable and must be replaced by a **ceiling** $\gamma\alpha/\lambda \leq \rho_{\max}$. This error was not detectable by re-reading: only the attempt to measure revealed it.

The value is method-dependent. With windows bounded by the return to baseline, their length ranges from 6 to 46 days; since attention decays more slowly than an exponential, a fit over a long window yields a smaller λ. The rank correlation reaches -0.94. A second, fixed-horizon estimator makes λ comparable across episodes; the median nonetheless varies by a factor of 1.7 across estimators, which forbids anchoring a regulatory threshold to a single value.

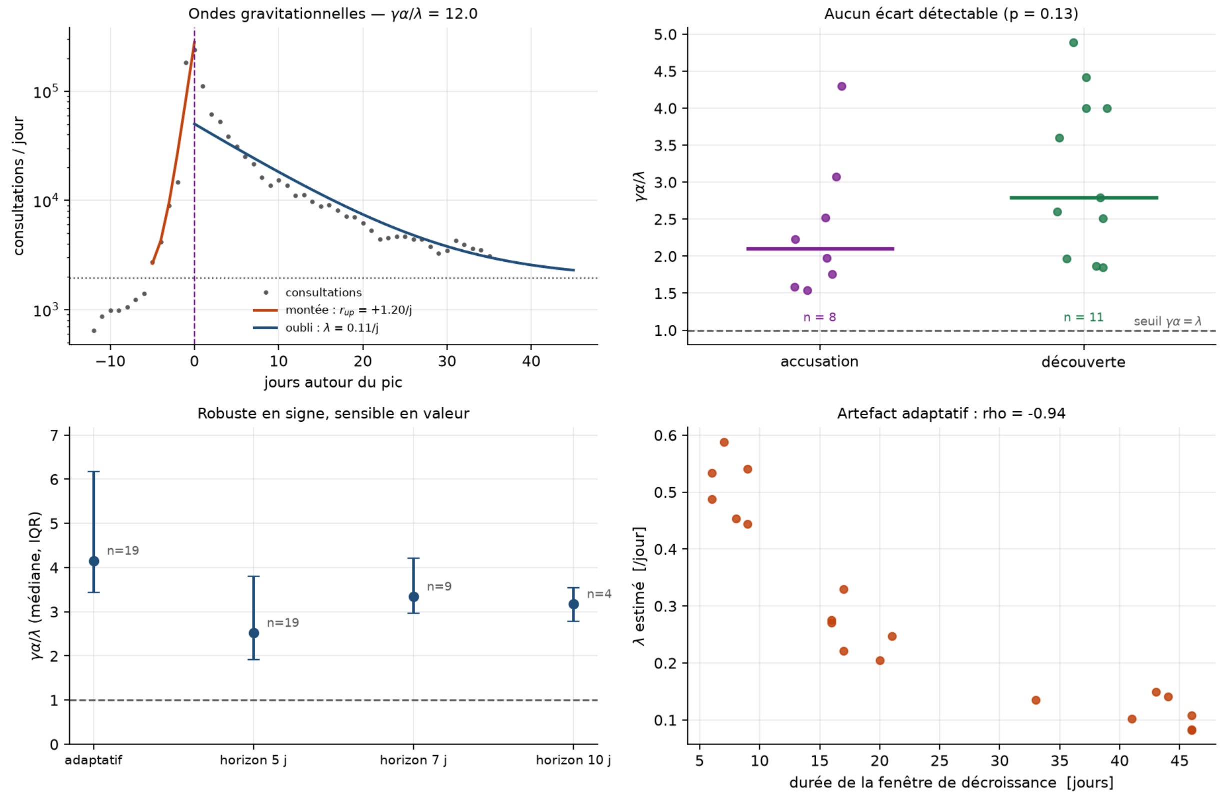

Figure 9: An attention episode and its two exponential regimes; the ratio distribution by emotional register; sensitivity of the median to the estimator; and the window artefact that forced the fixed-horizon estimator.

The emotional-charge mechanism is not supported. No detectable difference between the two registers ($p \geq 0.13$), and the point estimate goes the opposite way to the prediction. Sample sizes are small and the selection bias described below excludes the most charged cases, so it would be excessive to call this a refutation — but the mechanism cannot be presented as data-supported.

The method is blind to installed regimes. Eleven of twenty-four subjects yield no exploitable episode, and these are the archetypal cases: QAnon, health disinformation, vaccine hesitancy. Their attention does not form a spike but a *durable plateau*, which the rolling baseline absorbs — prominence-based detection does not see them. The procedure therefore selects against the very phenomenon the theory describes, which is the heaviest limitation of this calibration.

15 The founding analogy, and its verdict

This work started from an analogy: the larger a quantum system, the faster it decoheres; the larger a population, the harder agreement becomes. It produced measurable quantities — a diversity index, an amplification threshold, a critical temperature — and that is its merit. Its own claims were checked one by one, and none held.

What the analogy claimed	Verdict
The larger the system, the more disordered	**false, and backwards**: the retained diffusion term, $B(x) = k_B T(1-x^2)/N$, makes the macroscopic variable *more* deterministic at large N. Two objects were conflated — configuration entropy, extensive, and the noise of the macroscopic variable, in $1/N$
Decoherence increases entropy	**false**: a closed system evolves unitarily. It is the *reduced* subsystem's entropy that grows
$\tau_D \propto \tau_R/N$	a **heuristic** scaling law, not Zurek's result, which involves spatial separation and the thermal wavelength
"Social tunnelling"	**metaphor**: the correct analogue is activated barrier crossing [Kramers, 1940], which is classical
$1/k^N$ measures the improbability of consensus	describes a randomly drawn **initial state**, not a dynamics

To which is added an error that is not physical: the original account attributed fragmentation to the connectivity of "small-world" networks [Watts and Strogatz, 1998]. In mean field the Voter Model converges in $O(N)$: connectivity *accelerates* consensus. Fragmentation comes from the bias H_i and from homophily, not from topology alone.

What survives — free-energy landscape, phase transition, hysteresis, barrier crossing — belongs to classical statistical mechanics, in the Ising and sociophysics lineage [Galam, 2004, Castellano et al., 2009]. Nothing specifically quantum came through.

The analogy therefore worked as **scaffolding**: it suggested measurable quantities that outlive it. That is a real role, and it is not the same as being true. Calling it "fruitful hence sound" would be dishonest: it suggested measurements, and none of its claims survived verification. The index was renamed accordingly — it was the *Entropic Dissipation Index*, after decoherence, and is now the *Exposed Diversity Index*, after what it measures.

Part III
Caveats

16 Limitations

These are stated here rather than left to reviewers.

16.1 On the index and the standard

The retained form has never been measured on a real feed. It is defined (2), its cost is quantified by exhaustive enumeration, and that is all. The gap is one of *data*, not of method (§6), and §7 states what would close it.

The floor level cannot be deduced from the measurement. This work establishes the form of the standard and its price, not its value. Setting 0.60 rather than 0.40 is a political decision no computation presented here settles.

Discretisation into viewpoints is a political choice. Whoever defines the modalities defines the index. Measuring by divergence to a declared catalogue makes the choice explicit rather than burying it; it does not remove it.

The index remains gameable beyond what any automatic measure detects. The attacks of §3 are closed, but a platform can still serve formally divergent, substantively empty items — bin diversity without argument diversity. Any mandated metric becomes a target.

A floor is a constraint on what citizens see. Defensible, but an intervention in public debate, and presenting it as a neutral technical measure would be dishonest. Measuring individual feeds also raises a privacy question that §7 answers only in part, by aggregation.

16.2 On the exposure measurements

The form $e(R) = R^{-\eta}$ is posited. Only its severity is estimated. A cascade examination model, or one depending on the item, would produce a different rank dependence — which test (5) would detect just as well, since it only tests independence, but which the minimum detectable severity would no longer summarise.

Exhaustive enumeration bounds feed size. Eight positions over four viewpoints. Nothing guarantees that burial behaves the same on a fifty-item feed, and asserting it would require an approximate optimisation that would itself have to be shown not to invent the result.

The measured datasets are slices. A slice of Baidu-ULTR is not Baidu-ULTR; the Open Bandit Dataset measures a fashion recommendation on three horizontal thumbnails, and transporting its figures to a news feed is precisely the operation §6 warns against.

The ground-truth confrontation concerns a uniform target policy , the only one this dataset has a bucket for. Nothing guarantees importance weighting behaves as well for a target further from the logging policy — on the contrary, the effective sample size would be smaller still.

16.3 The two deepest assumptions, tested

Two assumptions run through this entire note without ever having been tested: that examination follows a power law, and that relevance is known. Neither is.

The exchangeability test survives a change of click model. Under a *cascade* model [Craswell et al., 2008] — the reader moves down the feed, stops on finding what they wanted, and otherwise continues with probability γ — test (5) rejects between $z = -208$ and $z = -241$ depending on γ, i.e. *more strongly* than on Baidu-ULTR. That is the result required: the test assumes nothing about the shape of attention, it tests only independence, and its negative verdict on MIND therefore holds under any model.

The power law does not survive. Under cascade, examination decays *geometrically* rather than polynomially. Simulation gives here the ground truth real data never provides — the positions actually examined — and the comparison is unequivocal: at rank twelve, the fitted $R^{-\hat{\eta}}$ overstates real exposure by a factor of 40 at $\gamma = 0.95$, 129 at $\gamma = 0.85$ and 4,110 at $\gamma = 0.60$. The fit nonetheless looks excellent, with a standard error of 0.04.

The consequence bears directly on §6: the severity $\hat{\eta} = 1.10 \pm 0.09$ measured on Baidu-ULTR means something *only if* examination there follows a power law — and Baidu-ULTR is a search results page, the cascade model's home ground. Severity must therefore be published **with its**

assumed shape, and that shape remains to be tested. The exchangeability test says whether the order carries information; it does not say in what form.

And estimated relevance erases the tuned methods' advantage. The comparison in §8.1 assumes relevance known. Having the methods rank on *perceived* relevance and judging them on the true one leaves their order unchanged — filter and MMR remain ahead — but their advantage over random selection falls from 16.4 points at zero noise to 5.8 points at a rank correlation of 0.50, and *inverts* below 0.36. The reason is mechanical: random selection is the only re-ranker that does not use relevance, hence the only one whose shortfall is invariant to noise.

At realistic relevance, the gap between re-rankers therefore matters far less than the quality of the underlying relevance engine — which reinforces, by a third route, the conclusion that this work's contribution is not its algorithm.

16.4 What the counter-expertise withdrew or restricted

The measuring instruments were confronted with the field's literature and with four counter-tests. Three results of this note come out modified.

A withdrawn conclusion. The comparison of the four diversity measures (§3) was carried out *at the same nominal floor*. Since the measures do not live on the same scale, that comparison is meaningless: at equal actually exposed diversity, the two retained measures cost the same, within 0.6% of the exact bound. The claim that proximity to a target "resists" burial better is a scale artefact. What makes the standard is the **level** required and **rank-awareness**, not the choice of measure.

A restricted conclusion. The figure "certified at 0.70, it exposes 0.36" assumes a $1/R$ attention discount, i.e. $\eta = 1$. Recomputed with the *measured* severity of a three-thumbnail banner ($\eta \approx 0.1$), the same feed exposes 0.747: burial disappears. At $\eta = 2$ it exposes only 0.157, and the loophole costs 2.7% instead of 20.8%. The attention discount must therefore be **measured on the surface**, and the stable price is that of the rank-aware standard — 20.8 to 24.1% depending on the surface.

A widened uncertainty. The estimator of §4 assumes a multiplicative click model. The literature documents an *affine* one, where readers click out of trust on irrelevant top-ranked items [Vardasbi et al., 2020], and proves that inverse propensity weighting cannot correct it. Under that model the estimated severity drifts by +12.8% without the standard error signalling it. Far less severe than positing η blindly, but the published interval is too narrow.

A confirmed recommendation. The doubly robust estimator does *worse* than plain weighting on the Open Bandit Dataset (−4.9% against +2.5%), and the effective sample size does not move: it depends only on the importance weights. No estimator replaces exploration.

And a caution from the literature. On Baidu-ULTR — the very dataset where this note measures $\hat{\eta} = 1.10$ — Hager et al. [2024] find that position-bias corrections improve click prediction *without* improving ranking quality as judged by expert annotators. The presence of the bias is confirmed by four concordant methods, which corroborates our figure; the next step does not follow mechanically.

16.5 On the model

Calibration barely begun. A single composite parameter is estimated (§14); J, T, γ and α taken separately have no estimation procedure. The most promising route would be inference of

Kramers-Moyal coefficients on long opinion series, which would allow the drift to be *tested* rather than fitted.

The model's one proper prediction is not verified. The emotional-charge mechanism α predicts a difference between registers — amplification, persistence, switching rate. Four independent measurements, including a replication across 440 subjects and a blind double-coded annotation, find no trace of it. The apparent gap in the pilot corpus was a selection artefact.

An analogy is not an explanation. Nothing here demonstrates that opinions *obey* statistical mechanics; the work establishes that a formalism borrowed from physics *reproduces* some of its behaviours. On the founding analogy itself, the verdict is sharper still (§15).

Individuals are not spins. Intentionality, irony and strategic anticonformism have no equivalent in a spin model; opinions are multidimensional [Deffuant et al., 2000]. Above all, the environment is not passive: an algorithm optimises a cost function, which makes it a strategic actor rather than a thermal bath. That is the analogy's weakest point — and it is also what makes the filter of §8 conceivable: a cost function can be changed.

Scope of the simulations. The regimes explored were chosen for legibility. No systematic sensitivity study was carried out, system sizes are modest, and the conclusions of Part II are qualitative: they concern the existence of regimes and the direction of dependencies, never transferable numerical values.

16.6 On the annotation

The coders are not independent. Blind double coding of the corpus yields a Fleiss κ of 0.921, but the three coders are instances of the same language model: their agreement necessarily overstates what independent human judges would produce. It is the last methodological reservation that no computation lifts.

17 Reproducibility

The entire work is openly available and runs in containers, with 556 automated tests. The structural checks cover Onsager's critical temperature (10), exact conservation of probability mass, the scaling exponents of consensus time, the strict positivity of the hysteresis loop area below T_c, the equivalence of filter (6) with an engagement filter at $\mu = 0$, the sign of the exchangeability test's rejection on a genuinely ordered log, and the exactness of the measurements recomputed from the aggregate tables of §7.

The impression logs used are not redistributed — they weigh 135 MB to 2.2 GB and each carries its own licence — but the repository keeps verified digests of them, which reproduce every figure published here identically. Every figure is regenerated by a notebook.

`https://github.com/s-geffroy/Indice-Diversite-Exposee`
`https://s-geffroy.github.io/Indice-Diversite-Exposee/`

This work is open to critical review. The most useful feedback would concern the retained form of the index and the level of any floor, or the measurement instruments of §4 and §5, where a methodological error would be the most costly.

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